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1 Completeness

We saw last time that even though there are no intervals of empty space between rational numbers, the rational number line $\mathbb{Q}$ has certain "holes". For example, we saw that there was no rational number x such that $x^2 = 2$, so $\mathbb{Q}$ has a hole at $\sqrt{2}$. To obtain the real number system $\mathbb{R}$, we fill in each such hole with an irrational number. One then says that the real numbers are *complete*, since the real number line has no such holes.

Completeness turns out to have profound consequences, and indeed the most important theorems of calculus follow from the completeness property of $\mathbb{R}$ (and are false for functions defined only on $\mathbb{Q}$). In order to see this, we will first need to turn our intuitive but vague idea of completeness above into a precise definition. To see how to do this, consider the following example. Let $\mathbb{Q}^+$ denote the set of positive rationals, and consider the subsets

$$A = \{p \in \mathbb{Q}^+ : p^2 < 2\}, \quad B = \{p \in \mathbb{Q}^+ : p^2 > 2\}.$$

Recall that the *maximum* of a set S of numbers is an element $M \in S$ such that $s \leq M$ for all $s \in S$. Similarly, the *minimum* of S is an element $m \in S$ such that $m \leq s$ for all $s \in S$.

Claim 1: The set A has no maximum and B has no minimum.

Similar to the definition of a maximum, if S is a subset of $\mathbb{R}$ (or $\mathbb{Q}$ or $\mathbb{Z}$), we say that an element $u \in \mathbb{R}$ is an *upper bound* for S, if $s \leq u$ for every $s \in S$. If S has an upper bound, then S is said to be *bounded above*. If u is an upper bound for S and $u \in S$, then $u = \max S$. However, an upper bound u for S need not be in S. Indeed, if u is an upper bound for S, then every number b such that $u \leq b$ is also an upper bound for S. Similar comments apply to a *lower bound* for S, which is a real number ℓ such that $\ell \leq s$ for every $s \in S$.

Claim 2: B is the set of upper bounds for A.

As we have noted, given an upper bound u for a set S, any b such that $u \leq b$ is also an upper bound for S. However, if $b < u$, then b need not be an upper bound for S. We can therefore ask, does a set S which is bounded above have a *least* upper bound? The set A in our example above is bounded above by every element of B. However, we have seen that B has no minimum element, so A has no least upper bound.

Suppose, instead, that we view A and B not as subsets of $\mathbb{Q}^+$, but as subsets of $\mathbb{R}^+$. Then, in addition to the elements of B, $\sqrt{2}$ is also a lower bound for A, since $a^2 < 2$ with a positive implies $a < \sqrt{2}$. Since $\sqrt{2} < b$ for every $b \in B$, $\sqrt{2}$ is the least upper bound for A.

This example shows that a hole in $\mathbb{Q}$ can result in a set which is bounded above, but which fails to have a least upper bound. In the example above, filling in the hole with the irrational number $\sqrt{2}$ provided a least upper bound for the set in $\mathbb{R}$. In a course on real analysis, you will construct $\mathbb{R}$ from $\mathbb{Q}$ and prove the following as a theorem, which we will take as an axiom:

Theorem. (Completeness of $\mathbb{R}$) Every nonempty subset of $\mathbb{R}$ which is bounded above has a least upper bound.

Remark. Note that the empty set is bounded above by every real number, and therefore has no least upper bound.

2 Suprema and infima

Definition. The least upper bound of a subset S of $\mathbb{R}$ is also called the *supremum* of S, and is denoted $\sup S$. It is characterized by the following two properties

(1) $s \leq \sup S$ for every $s \in S$ (which just says that $\sup S$ is an upper bound for S).

(2) If u is any upper bound for S, then $\sup S \leq u$ (which says that $\sup S$ is the *least* upper bound for S).

Proposition. If $\sup S$ exists, it is unique.

Proposition. If S has a maximum, then $\sup S = \max S$.

Instead of Property (2) above, it is often useful to instead replace it by its contrapositive (which is logically equivalent), which states that if $u < \sup S$, then u is not an upper bound for S.

Example. Consider the open interval

$$(a,b) := \{x \in \mathbb{R} : a < x < b\}.$$

By definition, b is an upper bound for (a,b). If $y < b$, then $y < \frac{y+b}{2} < b$, which shows that y is not an upper bound for (a,b). Hence, $\sup(a,b) = b$.

Remark. As we have seen, if S has a maximum, then $\sup S = \max S \in S$. The previous example shows that if S has no maximum but is bounded above, then $\sup S$ is not in S.

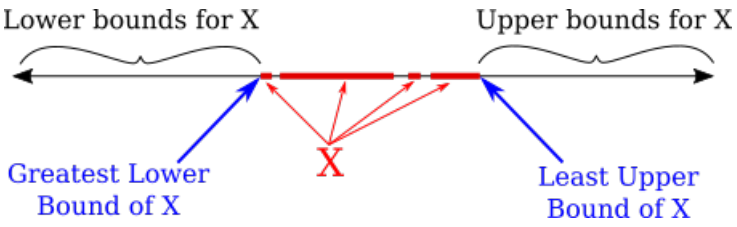
Definition. Similar to the definition of the least upper bound of S, if S is bounded below then one defines the *greatest lower bound* or *infimum* of S, denoted $\inf S$, by the properties

(1) $\inf S \leq s$ for every $s \in S$ (which just says that $\inf S$ is a lower bound for S)

(2) If ℓ is any lower bound for S, then $\ell \leq \inf S$ (which says that $\inf S$ is the *greatest lower bound* for S).

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The proof that $\inf S$ is unique, if it exists, is essentially the same as the proof for $\sup S$. Here is a potentially helpful picture:



Proposition: Every nonempty subset S of $\mathbb{R}$ which is bounded below has a greatest lower bound.

Proof. You will prove this in the first problem of the week. □

Exercise. Prove that if $\min S$ exists, then $\min S = \inf S$.

Exercise. Prove that $\inf(a,b) = a$.

3 Exercises

1. Let A and B be nonempty bounded subsets of $\mathbb{R}$ such that $B \subseteq A$. Prove that $\sup B \leq \sup A$.

2. Let $A \subseteq \mathbb{R}$ be bounded above and let $c \geq 0$. Define the set $cA = \{ca : a \in A\}$. Prove that

$$\sup(cA) = c \sup A.$$